Elastic's Role in Image Classification Pipelines

Elastic Agentic Talks

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Core Message

Before: External tools for vectors + custom orchestration.

After: Elastic acts as the unified engine for ingestion, vectorization, and image classification workflows.

Before Elastic-first Workflow

STEP	WHAT HAPPENED	PAIN POINT
1	Installed GitHub extension and integration stack	More setup and moving parts
2	Used Hugging Face model for image vectors	Endpoint and model operations overhead
3	Stored vectors and searched through separate components	Extra orchestration and maintenance

Previous Library: Elastiknn

How I used it

- Nearest-neighbor search before moving to Elastic-native vector search.
- Supported both **dense float vectors** and **sparse boolean vectors**.
- Helped run similarity search with exact and approximate approaches.

Credit

Special thanks to **Alex Klibisz** for building and maintaining Elastiknn.

Repository: github.com/alexklibisz/elastiknn

Project: Elasticsearch plugin for similarity / nearest-neighbor search.

Previous Library: img2vec

How I used it

- Generated image embeddings before Elastic-native inference flow.
- Used pre-trained PyTorch vision models for feature extraction.
- Indexed output vectors for similarity and semantic-style retrieval.

Credit

Special thanks to **Christian Safka** for building and sharing img2vec.

Repository: github.com/christiansafka/img2vec

Project: Use pre-trained PyTorch models to extract image embeddings.

Scale & Outcome

From my previous pipeline

What I built

- **img2vec** for image vectorization.
- **Elastiknn** for nearest-neighbor similarity search.
- Converted **50+ lakh images** into vectors.
- Injected vectorized data into Elasticsearch.

Result

Semantic search performed

very fast at large scale

This validated the approach and set up the transition story to an Elastic-native in-house workflow.

Elastic Vector Timeline

VERSION / PHASE	WHAT WAS AVAILABLE	DOCS REF
Before 8.0	Dense vectors existed, but **approximate kNN indexing was not supported**.	Before 8.0
8.0+	**Approximate kNN search added** for vector similarity at scale.	2026-04-27
9.0+	Platform matured with quantization options (int8_hnsw, bbq_hnsw, bfloat16).	2026-04-15
Inference Era	Endpoints support many models (ELSER v2, Jina embeddings) + rerankers.	2026-04-28

After Elastic-native Workflow

STEP	ELASTIC CAPABILITY	OUTCOME
1	**Elastic Cloud Serverless** for Search	Managed scaling and fast setup
2	**Inference endpoints** (Jina, ELSER, etc.)	No custom ML serving pipeline
3	**Vector fields** + semantic query	Single platform for ingest and retrieval

Elastic Agent Builder

What it gives you

- Build domain-specific AI agents on top of Elasticsearch data.
- Combine retrieval, tools, and orchestration in one workflow.
- Use vector search + filters + metadata for grounded responses.

Why it matters for this talk

- Same search foundation you used for image/vector retrieval.
- Natural next step from semantic search to agentic applications.
- Lets teams ship assistant-like experiences faster.

Agent Capabilities with Elastic

CAPABILITY	HOW ELASTIC HELPS	EXAMPLE USE CASE
Knowledge retrieval	Hybrid + semantic search over indexed content	Find visually similar products from text prompts
Grounded responses	Context from vectors, keywords, and metadata filters	Return answers tied to exact catalog documents
Reranking	Inference endpoints and reranker integrations	Improve top-k relevance before final answer
Production scale	Serverless deployment + managed scaling	High-volume image and multimodal search agents

Live Demo

- **Elastic Inference Service:** https://www.elastic.co/docs/explore-analyze/elastic-inference/eis
- **Jina endpoint API:** https://www.elastic.co/docs/api/doc/elasticsearch/operation/operation-inference-put-jinaai
- **Vector search quickstart:** https://www.elastic.co/docs/solutions/search/get-started/semantic-search
- **Cloud Serverless:** https://www.elastic.co/docs/deploy-manage/deploy/elastic-cloud/serverless

Project Links & References

My Repositories

- **kNN Image Search:** github.com/navinbalaji/elastic-knn-image-search
- **Elastic Copilot Search:** github.com/navinbalaji/elastic-copliot-search

External Libraries

- **img2vec:** github.com/christiansafka/img2vec
- **Elastiknn:** github.com/alexklibisz/elastiknn

Thank You!